Heat and material balance

SO3 Converter Ver 1 · SO3-V1-PE-HMB-001 · Rev 0

Department of Chemical Engineering, Chulalongkorn University (teaching)

REV	DATE	DESCRIPTION	ORIGINATOR	CHECKED / APPROVED
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Generated for teaching from the SO3 Converter Ver 1 operator training simulator. Every number comes from the simulator model (design point, noise off) or is marked assumed. Not for construction. No client, contractor or vendor is represented.

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1 · Purpose and scope

This document issues the heat and material balance (HMB) of the SO_2 converter section of SO3 Converter Ver 1 (sulfur-burning H_2SO_4 plant, 1000 kt/y): one column per stream of PFD-201, in the design case and two off-design cases run on the simulator model. It also gives the balances and their closure, an energy summary per equipment item, the line list with velocities and pressure drops, blower and pump hydraulics, design margins, the checks against the model and the other documents, and the engineering queries raised.

Battery limits: dried air in, molten sulfur in, circulating acid in and out, steam out, tail gas to the stack. The drying tower, acid coolers, product acid system and steam drum are outside the model; their numbers are estimates. Related documents: PFD-201 (PFD tab), SO3-PID-001/002 rev A (P&ID tab), SO3-V1-ME-EQL-001 (equipment list), SO3-V1-PE-RLB-001 (relief load basis), SO3-V1-SA-HZP-001 (HAZOP report).

2 · Design basis

Item	Basis	Source
Capacity	1000 kt/y H_2SO_4 (100 %) at 8000 h/y = 125 t/h; model FI-309 124.99 t/h	model
Sulfur feed	40.94 t/h molten sulfur at 140 °C (assumed), 1277 kmol/h	model / assumed T
Air	238.5 kNm³/h dried air (combustion 204.4, dilution 34.1), 307.0 t/h at 90 °C	model
Converter feed	12.0 % SO_2, 9.0 % O_2, 420 °C, PI-200 35 kPa g	model
Conversion, emissions	XI-308 99.80 %; stack SO_2 292 ppm (permit 500 ppm); acid mist 10.5 mg/Nm³	model
Acid strengths	circulating 98.5 wt % H_2SO_4 (AI-303); leaving T-301 H_2SO_4 99.29 wt %, H_2O rest; leaving T-302 H_2SO_4 98.55 wt %, H_2O rest; product 98.5 %	model / HMB
Ambient	ambient 30 °C (K.Tamb); air at the blower 90 °C (K.Tair); hot-day case +20 K	model
HP steam	94.0 t/h at 4.0 MPa g, 292 °C	model
Boiler feed water	150 t/h at 105 °C to E-203/E-204/E-205	model
Cooling water	acid coolers outside the model; duty ≈ 63 MW (absorber heat, HMB)	estimate
Power	K-101 4200 kW (econ); pumps in section 6	model / estimate

2.1 Cases

Case	Basis	S, t/h	Air, kNm³/h	PI-101	PI-200	FV-101 %	TI-203 °C	Stack ppm	Acid t/h	Steam t/h	K-101 kW
Design	FIC-103 40.94 t/h, AIC-104 12.0 % (snapDesign)	40.94	238.5	60.0	35.0	54.9	631.3	292	125.0	94.0	4200
110 % rate	FIC-103 SP 45.0 t/h, loops AUTO, 4800 s, trips off	45.00	262.2	56.9	42.3	97.1	632.3	343	137.3	103.8	4840
Turndown	FIC-103 SP 28.0 t/h, loops AUTO, 4800 s, trips off	28.00	163.1	68.0	16.4	23.4	628.6	185	85.5	63.0	2375

Off-design cases are steady states of the model: 4800 s at dt 1 s from the design state with the sulfur set point changed, every loop in its design mode, trips switched off so that limits show (script tabs/eproc/tools/gen.js).

3 · Heat and material balance — Design case

FIC-103 40.94 t/h, AIC-104 12.0 % (snapDesign). Stream numbers of PFD-201. * pressure estimated from the model resistances (only PI-101, PI-200 and the sulfur supply are model values). Gas mol %, liquid wt %.

Stream	1	2	3	4	5	6	7	8	9	10	11
Description	Dried air to K-101	K-101 discharge	Combustion air, FV-101	Dilution air, FV-102	Molten sulfur to P-101A/B	Sulfur to burner, FV-103	Furnace gas	Gas into B-101	Hot-gas bypass TV-201	Gas leaving B-101	After bypass joins
BFD no.	1	1	2	3	4	4	5	—	—	—	6
Phase	gas	gas	gas	gas	liquid	liquid	gas	gas	gas	gas	gas
Mass flow, t/h	307.01	307.01	263.17	43.85	40.94	40.94	304.12	241.06	63.05	241.06	304.12
Molar flow, kmol/h	10641.5	10641.5	9121.7	1519.8	1277.0	1277.0	9121.7	7230.5	1891.3	7230.5	9121.7
kNm³/h (gas) | m³/h (liq.)	238.52	238.52	204.45	34.06	22.9	22.9	204.45	162.06	42.39	162.06	204.45
Actual flow, m³/s	88.1	55.3	53.8	9.3	—	—	209.2	165.9	43.4	67.1	114.7
Temperature, °C	90.0	90.0	90.0	90.0	140.0	140.0	1120.6	1120.6	1120.6	274.4	468.7
Pressure, kPa g	0.0*	60.0	41.0*	35.0*	10.0*	600.0	39.0*	39.0*	39.0*	35.0*	35.0*
SO_2, mol %	0.00	0.00	0.00	0.00	—	—	14.00	14.00	14.00	14.00	14.00
O_2, mol %	21.00	21.00	21.00	21.00	—	—	7.00	7.00	7.00	7.00	7.00
N_2, mol %	79.00	79.00	79.00	79.00	—	—	79.00	79.00	79.00	79.00	79.00
SO_3, mol %	0.00	0.00	0.00	0.00	—	—	0.00	0.00	0.00	0.00	0.00
Liquid, wt %	—	—	—	—	S 100 % (molten)	S 100 % (molten)	—	—	—	—	—
Density, kg/m³	0.968	1.542	1.360	1.303	1790	1790	0.404	0.404	0.404	0.998	0.737

Stream	12	13	14	15	16	17	18	19	20	21	22
Description	After dilution air	Converter feed, bed 1	Start-up line HV-112	Bed 1 exit	SH-201 to E-201	E-201 to bed 2	Bed 2 exit	E-202 to bed 3	Bed 3 exit	Gas to T-301	Gas leaving T-301
BFD no.	7	7	21	—	—	—	—	—	8	9	10
Phase	gas	gas	gas	gas	gas	gas	gas	gas	gas	gas	gas
Mass flow, t/h	347.96	347.96	0.00	347.96	347.96	347.96	347.96	347.96	347.96	347.96	250.03
Molar flow, kmol/h	10641.5	10641.5	0.0	10238.7	10238.7	10238.7	10079.1	10079.1	10029.6	10029.6	8806.4
kNm³/h (gas) | m³/h (liq.)	238.52	238.52	0.00	229.49	229.49	229.49	225.91	225.91	224.80	224.80	197.39
Actual flow, m³/s	125.0	125.0	0.0	162.3	159.9	131.6	147.8	133.2	140.8	89.3	61.9
Temperature, °C	420.0	420.0	420.0	631.3	610.3	445.0	528.0	440.0	465.3	190.0	83.0
Pressure, kPa g	35.0*	35.0	35.0*	30.5*	29.3*	27.8*	24.9*	23.3*	20.2*	18.8*	15.7*
SO_2, mol %	12.00	12.00	0.00	4.60	4.60	4.60	1.51	1.51	0.53	0.53	0.60
O_2, mol %	9.00	9.00	0.00	5.42	5.42	5.42	3.92	3.92	3.45	3.45	3.93
N_2, mol %	79.00	79.00	0.00	82.11	82.11	82.11	83.41	83.41	83.82	83.82	95.46
SO_3, mol %	0.00	0.00	0.00	7.87	7.87	7.87	11.16	11.16	12.20	12.20	0.01
Liquid, wt %	—	—	—	—	—	—	—	—	—	—	—
Density, kg/m³	0.773	0.773	0.685	0.596	0.604	0.735	0.654	0.726	0.687	1.083	1.122

Numbers from the simulator model, design point, noise off; "assumed" where stated.

Stream	23	24	25	26	27	28	29	30	31	32	33
Description	Return E-202→E-201	Return gas to bed 4	Bed 4 exit	E-204 to bed 5	Bed 5 exit	Gas to T-302	Tail gas to stack	Acid to T-301	Acid from T-301	Acid to T-302	Acid from T-302
BFD no.	—	11	—	—	12	13	14	—	15	—	16
Phase	gas	gas	gas	gas	gas	gas	gas	liquid	liquid	liquid	liquid
Mass flow, t/h	250.03	250.03	250.03	250.03	250.03	250.03	245.93	2880.00	2977.93	1800.00	1804.10
Molar flow, kmol/h	8806.4	8806.4	8781.8	8781.8	8781.1	8781.1	8729.9	31321.3	31321.3	19575.8	19575.8
kNm³/h (gas) | m³/h (liq.)	197.39	197.39	196.83	196.83	196.82	196.82	195.67	1600.0	1654.4	1000.0	1002.3
Actual flow, m³/s	84.0	124.6	130.0	129.0	131.9	82.6	69.0	—	—	—	—
Temperature, °C	203.6	425.0	441.0	428.0	427.4	160.0	80.0	80.0	124.5	80.0	90.6
Pressure, kPa g	14.2*	12.6*	10.1*	8.9*	6.4*	5.0*	1.9*	488.5*	0.0*	488.5*	0.0*
SO_2, mol %	0.60	0.60	0.04	0.04	0.03	0.03	0.03	—	—	—	—
O_2, mol %	3.93	3.93	3.66	3.66	3.65	3.65	3.67	—	—	—	—
N_2, mol %	95.46	95.46	95.73	95.73	95.74	95.74	96.30	—	—	—	—
SO_3, mol %	0.01	0.01	0.57	0.57	0.58	0.58	0.00	—	—	—	—
Liquid, wt %	—	—	—	—	—	—	—	H_2SO_4 98.50 wt %	H_2SO_4 99.29 wt %	H_2SO_4 98.50 wt %	H_2SO_4 98.55 wt %
Density, kg/m³	0.827	0.557	0.534	0.538	0.527	0.841	0.991	1800	1800	1800	1800

4 · Heat and material balance — 110 % rate

FIC-103 SP 45.0 t/h, loops AUTO, 4800 s, trips off. Stream numbers of PFD-201. * pressure estimated from the model resistances (only PI-101, PI-200 and the sulfur supply are model values). Gas mol %, liquid wt %.

Stream	1	2	3	4	5	6	7	8	9	10	11
Description	Dried air to K-101	K-101 discharge	Combustion air, FV-101	Dilution air, FV-102	Molten sulfur to P-101A/B	Sulfur to burner, FV-103	Furnace gas	Gas into B-101	Hot-gas bypass TV-201	Gas leaving B-101	After bypass joins
BFD no.	1	1	2	3	4	4	5	—	—	—	6
Phase	gas	gas	gas	gas	liquid	liquid	gas	gas	gas	gas	gas
Mass flow, t/h	337.46	337.46	289.27	48.20	45.00	45.00	334.27	265.73	68.54	265.73	334.27
Molar flow, kmol/h	11696.8	11696.8	10026.3	1670.5	1403.6	1403.6	10026.3	7970.4	2055.9	7970.4	10026.3
kNm³/h (gas) | m³/h (liq.)	262.17	262.17	224.73	37.44	25.1	25.1	224.73	178.65	46.08	178.65	224.73
Actual flow, m³/s	96.8	62.0	55.7	9.8	—	—	217.9	173.2	44.7	70.4	119.6
Temperature, °C	90.0	90.0	90.0	90.0	140.0	140.0	1123.5	1123.5	1123.5	276.2	468.7
Pressure, kPa g	0.0*	56.9	49.5*	42.3*	10.0*	600.0	47.1*	47.1*	47.1*	42.3*	42.3*
SO_2, mol %	0.00	0.00	0.00	0.00	—	—	14.00	14.00	14.00	14.00	14.00
O_2, mol %	21.00	21.00	21.00	21.00	—	—	7.00	7.00	7.00	7.00	7.00
N_2, mol %	79.00	79.00	79.00	79.00	—	—	79.00	79.00	79.00	79.00	79.00
SO_3, mol %	0.00	0.00	0.00	0.00	—	—	0.00	0.00	0.00	0.00	0.00
Liquid, wt %	—	—	—	—	S 100 % (molten)	S 100 % (molten)	—	—	—	—	—
Density, kg/m³	0.968	1.512	1.441	1.372	1790	1790	0.426	0.426	0.426	1.048	0.776

Stream	12	13	14	15	16	17	18	19	20	21	22
Description	After dilution air	Converter feed, bed 1	Start-up line HV-112	Bed 1 exit	SH-201 to E-201	E-201 to bed 2	Bed 2 exit	E-202 to bed 3	Bed 3 exit	Gas to T-301	Gas leaving T-301
BFD no.	7	7	21	—	—	—	—	—	8	9	10
Phase	gas	gas	gas	gas	gas	gas	gas	gas	gas	gas	gas
Mass flow, t/h	382.47	382.47	0.00	382.47	382.47	382.47	382.47	382.47	382.47	382.47	274.71
Molar flow, kmol/h	11696.8	11696.8	0.0	11252.2	11252.2	11252.2	11076.9	11076.9	11023.5	11023.5	9677.5
kNm³/h (gas) | m³/h (liq.)	262.17	262.17	0.00	252.21	252.21	252.21	248.28	248.28	247.08	247.08	216.91
Actual flow, m³/s	130.4	130.4	0.0	170.3	167.9	138.4	156.0	140.9	149.5	95.0	66.2
Temperature, °C	420.0	420.0	420.0	632.3	610.1	445.0	528.0	440.0	464.9	190.0	83.0
Pressure, kPa g	42.3*	42.3	42.3*	36.8*	35.4*	33.5*	30.1*	28.2*	24.4*	22.7*	19.0*
SO_2, mol %	12.00	12.00	0.00	4.57	4.57	4.57	1.48	1.48	0.52	0.52	0.59
O_2, mol %	9.00	9.00	0.00	5.40	5.40	5.40	3.91	3.91	3.44	3.44	3.92
N_2, mol %	79.00	79.00	0.00	82.12	82.12	82.12	83.42	83.42	83.83	83.83	95.48
SO_3, mol %	0.00	0.00	0.00	7.90	7.90	7.90	11.19	11.19	12.22	12.22	0.01
Liquid, wt %	—	—	—	—	—	—	—	—	—	—	—
Density, kg/m³	0.815	0.815	0.722	0.624	0.633	0.768	0.681	0.754	0.711	1.118	1.153

Stream	23	24	25	26	27	28	29	30	31	32	33
Description	Return E-202→E-201	Return gas to bed 4	Bed 4 exit	E-204 to bed 5	Bed 5 exit	Gas to T-302	Tail gas to stack	Acid to T-301	Acid from T-301	Acid to T-302	Acid from T-302
BFD no.	—	11	—	—	12	13	14	—	15	—	16
Phase	gas	gas	gas	gas	gas	gas	gas	liquid	liquid	liquid	liquid
Mass flow, t/h	274.71	274.71	274.71	274.71	274.71	274.71	270.36	2880.00	2987.76	1800.00	1804.35
Molar flow, kmol/h	9677.5	9677.5	9651.5	9651.5	9650.7	9650.7	9596.4	31321.3	31321.3	19575.8	19575.8
kNm³/h (gas) | m³/h (liq.)	216.91	216.91	216.33	216.33	216.31	216.31	215.09	1600.0	1659.9	1000.0	1002.4
Actual flow, m³/s	90.0	133.9	140.2	139.4	143.2	91.1	75.5	—	—	—	—
Temperature, °C	203.7	425.0	440.4	428.0	427.6	165.7	80.0	80.0	128.8	80.0	92.1
Pressure, kPa g	17.1*	15.2*	12.2*	10.8*	7.7*	6.1*	2.3*	488.5*	0.0*	488.5*	0.0*
SO_2, mol %	0.59	0.59	0.05	0.05	0.03	0.03	0.03	—	—	—	—
O_2, mol %	3.92	3.92	3.66	3.66	3.65	3.65	3.67	—	—	—	—
N_2, mol %	95.48	95.48	95.74	95.74	95.75	95.75	96.29	—	—	—	—
SO_3, mol %	0.01	0.01	0.55	0.55	0.56	0.56	0.00	—	—	—	—
Liquid, wt %	—	—	—	—	—	—	—	H_2SO_4 98.50 wt %	H_2SO_4 99.37 wt %	H_2SO_4 98.50 wt %	H_2SO_4 98.56 wt %
Density, kg/m³	0.848	0.570	0.544	0.547	0.533	0.838	0.995	1800	1800	1800	1800

5 · Heat and material balance — Turndown

FIC-103 SP 28.0 t/h, loops AUTO, 4800 s, trips off. Stream numbers of PFD-201. * pressure estimated from the model resistances (only PI-101, PI-200 and the sulfur supply are model values). Gas mol %, liquid wt %.

Stream	1	2	3	4	5	6	7	8	9	10	11
Description	Dried air to K-101	K-101 discharge	Combustion air, FV-101	Dilution air, FV-102	Molten sulfur to P-101A/B	Sulfur to burner, FV-103	Furnace gas	Gas into B-101	Hot-gas bypass TV-201	Gas leaving B-101	After bypass joins
BFD no.	1	1	2	3	4	4	5	—	—	—	6
Phase	gas	gas	gas	gas	liquid	liquid	gas	gas	gas	gas	gas
Mass flow, t/h	209.98	209.98	179.99	29.99	28.00	28.00	207.99	163.03	44.97	163.03	207.99
Molar flow, kmol/h	7278.0	7278.0	6238.6	1039.4	873.4	873.4	6238.6	4889.8	1348.8	4889.8	6238.6
kNm³/h (gas) | m³/h (liq.)	163.13	163.13	139.83	23.30	15.6	15.6	139.83	109.60	30.23	109.60	139.83
Actual flow, m³/s	60.2	36.0	43.4	7.4	—	—	166.4	130.4	36.0	51.9	90.8
Temperature, °C	90.0	90.0	90.0	90.0	140.0	140.0	1107.7	1107.7	1107.7	268.0	468.7
Pressure, kPa g	0.0*	68.0	19.2*	16.4*	10.0*	600.0	18.2*	18.2*	18.2*	16.4*	16.4*
SO_2, mol %	0.00	0.00	0.00	0.00	—	—	14.00	14.00	14.00	14.00	14.00
O_2, mol %	21.00	21.00	21.00	21.00	—	—	7.00	7.00	7.00	7.00	7.00
N_2, mol %	79.00	79.00	79.00	79.00	—	—	79.00	79.00	79.00	79.00	79.00
SO_3, mol %	0.00	0.00	0.00	0.00	—	—	0.00	0.00	0.00	0.00	0.00
Liquid, wt %	—	—	—	—	S 100 % (molten)	S 100 % (molten)	—	—	—	—	—
Density, kg/m³	0.968	1.618	1.151	1.125	1790	1790	0.347	0.347	0.347	0.872	0.636

Stream	12	13	14	15	16	17	18	19	20	21	22
Description	After dilution air	Converter feed, bed 1	Start-up line HV-112	Bed 1 exit	SH-201 to E-201	E-201 to bed 2	Bed 2 exit	E-202 to bed 3	Bed 3 exit	Gas to T-301	Gas leaving T-301
BFD no.	7	7	21	—	—	—	—	—	8	9	10
Phase	gas	gas	gas	gas	gas	gas	gas	gas	gas	gas	gas
Mass flow, t/h	237.98	237.98	0.00	237.98	237.98	237.98	237.98	237.98	237.98	237.98	171.23
Molar flow, kmol/h	7278.0	7278.0	0.0	7005.8	7005.8	7005.8	6896.4	6896.4	6861.0	6861.0	6027.3
kNm³/h (gas) | m³/h (liq.)	163.13	163.13	0.00	157.03	157.03	157.03	154.58	154.58	153.78	153.78	135.10
Actual flow, m³/s	99.0	99.0	0.0	126.2	124.4	101.7	112.9	101.2	105.8	66.6	45.6
Temperature, °C	420.0	420.0	420.0	628.6	611.1	445.0	527.8	440.0	466.1	190.0	83.0
Pressure, kPa g	16.4*	16.4	16.4*	14.3*	13.7*	13.0*	11.7*	10.9*	9.4*	8.8*	7.4*
SO_2, mol %	12.00	12.00	0.00	4.69	4.69	4.69	1.60	1.60	0.57	0.57	0.65
O_2, mol %	9.00	9.00	0.00	5.46	5.46	5.46	3.96	3.96	3.47	3.47	3.95
N_2, mol %	79.00	79.00	0.00	82.07	82.07	82.07	83.37	83.37	83.80	83.80	95.39
SO_3, mol %	0.00	0.00	0.00	7.77	7.77	7.77	11.07	11.07	12.16	12.16	0.01
Liquid, wt %	—	—	—	—	—	—	—	—	—	—	—
Density, kg/m³	0.668	0.668	0.591	0.524	0.531	0.650	0.585	0.653	0.625	0.992	1.043

Stream	23	24	25	26	27	28	29	30	31	32	33
Description	Return E-202→E-201	Return gas to bed 4	Bed 4 exit	E-204 to bed 5	Bed 5 exit	Gas to T-302	Tail gas to stack	Acid to T-301	Acid from T-301	Acid to T-302	Acid from T-302
BFD no.	—	11	—	—	12	13	14	—	15	—	16
Phase	gas	gas	gas	gas	gas	gas	gas	liquid	liquid	liquid	liquid
Mass flow, t/h	171.23	171.23	171.23	171.23	171.23	171.23	168.15	2880.00	2946.75	1800.00	1803.09
Molar flow, kmol/h	6027.3	6027.3	6008.4	6008.4	6008.2	6008.2	5969.6	31321.3	31321.3	19575.8	19575.8
kNm³/h (gas) | m³/h (liq.)	135.10	135.10	134.67	134.67	134.67	134.67	133.80	1600.0	1637.1	1000.0	1001.7
Actual flow, m³/s	61.4	90.6	93.7	92.2	93.1	55.4	47.6	—	—	—	—
Temperature, °C	203.0	425.0	442.5	428.0	426.7	140.7	80.0	80.0	110.7	80.0	86.1
Pressure, kPa g	6.6*	5.9*	4.7*	4.2*	3.0*	2.4*	0.9*	488.5*	0.0*	488.5*	0.0*
SO_2, mol %	0.65	0.65	0.03	0.03	0.02	0.02	0.02	—	—	—	—
O_2, mol %	3.95	3.95	3.65	3.65	3.64	3.64	3.67	—	—	—	—
N_2, mol %	95.39	95.39	95.69	95.69	95.70	95.70	96.31	—	—	—	—
SO_3, mol %	0.01	0.01	0.63	0.63	0.64	0.64	0.00	—	—	—	—
Liquid, wt %	—	—	—	—	—	—	—	H_2SO_4 98.50 wt %	H_2SO_4 99.04 wt %	H_2SO_4 98.50 wt %	H_2SO_4 98.54 wt %
Density, kg/m³	0.775	0.525	0.508	0.516	0.511	0.859	0.981	1800	1800	1800	1800

Numbers from the simulator model, design point, noise off; "assumed" where stated.

6 · Balances, closure and energy summary

Balances — Design

Balance	Equation	Unit	In / left	Out / right	Closure, %
Overall mass (gas side)	Air (1) + sulfur (6) = tail gas (29) + SO_3 absorbed in T-301 and T-302	t/h	347.954	347.962	0.0022
Sulfur	S in (FIC-103) = S in acid (FI-309) + S at the stack (SO_2 + SO_3 mist)	mol/s	354.717	354.717	0.0000
Nitrogen	N_2 in the air (1) = N_2 in the tail gas (29)	kmol/h	8406.789	8406.789	0.0000
Acid make	H_2SO_4 from SO_3 absorbed (HMB) = FI-309	t/h	124.993	124.993	0.0000
IPA acid circuit	Acid in (30) + SO_3 absorbed = acid out (31)	t/h	2977.931	2977.931	0.0000
Gas volume	Air (1) − 1.5 × S to acid = tail gas (29)	kNm³/h	195.672	195.672	−0.0001

Balances — 110 % rate

Balance	Equation	Unit	In / left	Out / right	Closure, %
Overall mass (gas side)	Air (1) + sulfur (6) = tail gas (29) + SO_3 absorbed in T-301 and T-302	t/h	382.461	382.469	0.0022
Sulfur	S in (FIC-103) = S in acid (FI-309) + S at the stack (SO_2 + SO_3 mist)	mol/s	389.894	389.894	0.0000
Nitrogen	N_2 in the air (1) = N_2 in the tail gas (29)	kmol/h	9240.486	9240.486	0.0000
Acid make	H_2SO_4 from SO_3 absorbed (HMB) = FI-309	t/h	137.342	137.342	0.0000
IPA acid circuit	Acid in (30) + SO_3 absorbed = acid out (31)	t/h	2987.762	2987.762	0.0000
Gas volume	Air (1) − 1.5 × S to acid = tail gas (29)	kNm³/h	215.093	215.093	−0.0001

Balances — Turndown

Balance	Equation	Unit	In / left	Out / right	Closure, %
Overall mass (gas side)	Air (1) + sulfur (6) = tail gas (29) + SO_3 absorbed in T-301 and T-302	t/h	237.976	237.981	0.0022
Sulfur	S in (FIC-103) = S in acid (FI-309) + S at the stack (SO_2 + SO_3 mist)	mol/s	242.601	242.601	0.0000
Nitrogen	N_2 in the air (1) = N_2 in the tail gas (29)	kmol/h	5749.636	5749.636	0.0000
Acid make	H_2SO_4 from SO_3 absorbed (HMB) = FI-309	t/h	85.549	85.549	0.0000
IPA acid circuit	Acid in (30) + SO_3 absorbed = acid out (31)	t/h	2946.748	2946.748	0.0000
Gas volume	Air (1) − 1.5 × S to acid = tail gas (29)	kNm³/h	133.804	133.804	−0.0002

Energy summary per equipment, MW

Item	Equipment	Basis	Design	110 % rate	Turndown	Source
K-101	Main air blower	shaft power (econ: 4200·(F/Fdes)^1.5 kW)	4.20	4.84	2.38	model
F-101	Sulfur furnace	sulfur burning S × K.DHS 295 kJ/mol	104.64	115.02	71.57	model
B-101	Waste heat boiler	gas side ṁcp(TI-106 − TI-107)	61.64	68.06	41.31	model
B-101	Waste heat boiler	steam FI-108 × (2801 − 440) kJ/kg (model basis)	61.64	68.06	41.31	model
B-101	Waste heat boiler	steam FI-108 × (2801 − 1087) kJ/kg (boiling only, Process tab)	44.75	49.41	29.99	model
R-201	Bed 1	Sfeed × ΔX × 98.5 kJ/mol	22.04	24.33	14.90	model
R-201	Bed 2	same	8.73	9.59	5.99	model
R-201	Bed 3	same	2.71	2.92	1.94	model
R-201	Bed 4	same	1.35	1.42	1.03	model
R-201	Bed 5	same	0.04	0.04	0.01	model
SH-201	Steam superheater	P.qSH (gas → steam)	2.24	2.59	1.27	model
E-201	Hot interpass exchanger	P.q201 (gas → return gas)	17.24	18.93	11.83	model
E-202	Cold interpass exchanger	P.q202 (gas → return gas)	9.16	10.07	6.24	model
E-203	Economizer (IPA)	BFW C × (Tout − 105 °C)	27.70	30.42	18.97	model
E-204	Economizer (bed 5)	same	1.03	1.08	0.79	model
E-205	Economizer (FAT)	same	20.77	22.36	15.18	model
T-301	Intermediate absorber	gas cooling TI-251 → TI-305 + SO_3 absorption (132 kJ/mol, assumed)	55.24	60.78	37.68	estimate
T-302	Final absorber	gas cooling TI-252 → 80 °C (assumed) + SO_3 absorption	7.93	9.12	4.55	estimate
	Released	sulfur burning (K.DHS) + SO_2 oxidation	139.5	153.3	95.4	model
	Recovered	B-101 + SH-201 + E-203/E-204/E-205	113.4	124.5	77.5	model

Released at design 139.5 MW (sulfur 104.6 with K.DHS 295 kJ/mol; with 296.8 kJ/mol 105.3); recovered 113.4 MW = 81 %. Absorber duties include the SO_3 absorption heat (132 kJ/mol, assumed), which is not in the model and goes to the acid coolers.

7 · Line list and line sizing

Each line is sized on the largest flow of the three cases (110 % rate): the smallest standard size whose velocity stays at or below the upper criterion. Criteria are typical practice: gas and air ducts 15–25 m/s, acid pump discharge 1–2 m/s, gravity acid return ≤ 1 m/s, molten sulfur ≤ 1 m/s, BFW 1–3 m/s, HP steam 20–40 m/s. ΔP by Darcy–Weisbach (Swamee–Jain); roughness 0.15 mm ducts / 0.05 mm pipes; gas viscosity by Sutherland for air (assumed). Ducts in 100 mm steps, pipes DN with standard-wall ID.

Line	Str.	Svc	Service	Size	T °C	P kPa g	ṁ t/h	ρ kg/m³	v des m/s	v 110 %	v turn.	Re	ΔP kPa/100 m	Criterion m/s	OK
L-002	2	PA	K-101 discharge duct	Ø 1.8 m	90	60	307.0	1.542	21.74	24.37	14.17	2.8e+06	0.250	15–25	✓
L-003	3	PA	Combustion air to F-101	Ø 1.7 m	90	41	263.2	1.360	23.68	24.56	19.13	2.6e+06	0.280	15–25	✓
L-004	4	PA	Dilution air	Ø 0.8 m	90	35	43.9	1.303	18.60	19.41	14.74	9.1e+05	0.415	15–25	✓
L-007	7	PG	Furnace gas to B-101 (refractory lined)	Ø 3.4 m	1121	39	304.1	0.404	23.05	24.00	18.33	6.3e+05	0.042	15–25	✓
L-009	9	PG	B-101 hot-gas bypass TV-201 (refractory lined)	Ø 1.6 m	1121	39	63.0	0.404	21.58	22.22	17.89	2.8e+05	0.091	15–25	✓
L-013	13	PG	Converter feed to bed 1	Ø 2.6 m	420	35	348.0	0.773	23.54	24.56	18.65	1.4e+06	0.102	15–25	✓
L-015	15	PG	Bed 1 exit to SH-201	Ø 3.0 m	631	30	348.0	0.596	22.96	24.10	17.86	1e+06	0.066	15–25	✓
L-017	17	PG	E-201 to bed 2	Ø 2.7 m	445	28	348.0	0.735	22.98	24.17	17.75	1.3e+06	0.089	15–25	✓
L-018	18	PG	Bed 2 exit to E-202	Ø 2.9 m	528	25	348.0	0.654	22.37	23.61	17.10	1.2e+06	0.071	15–25	✓
L-019	19	PG	E-202 to bed 3	Ø 2.7 m	440	23	348.0	0.726	23.26	24.60	17.67	1.4e+06	0.090	15–25	✓
L-020	20	PG	Bed 3 exit to E-203	Ø 2.8 m	465	20	348.0	0.687	22.86	24.27	17.17	1.3e+06	0.080	15–25	✓
L-021	21	PG	E-203 to T-301	Ø 2.3 m	190	19	348.0	1.083	21.49	22.87	16.04	2.1e+06	0.132	15–25	✓
L-022	22	PG	T-301 exit to E-202 (cold, wet-acid service)	Ø 1.9 m	83	16	250.0	1.122	21.83	23.33	16.09	2.2e+06	0.175	15–25	✓
L-024	24	PG	E-201 to bed 4	Ø 2.7 m	425	13	250.0	0.557	21.77	23.38	15.83	9.8e+05	0.063	15–25	✓
L-025	25	PG	Bed 4 exit to E-204	Ø 2.7 m	441	10	250.0	0.534	22.71	24.48	16.36	9.7e+05	0.065	15–25	✓
L-027	27	PG	Bed 5 exit to E-205	Ø 2.8 m	427	6	250.0	0.527	21.42	23.26	15.12	9.5e+05	0.055	15–25	✓
L-028	28	PG	E-205 to T-302	Ø 2.2 m	160	5	250.0	0.841	21.73	23.95	14.57	1.7e+06	0.112	15–25	✓
L-029	29	PG	T-302 to stack	Ø 2.0 m	80	2	245.9	0.991	21.95	24.03	15.16	2.1e+06	0.148	15–25	✓
L-005	5	SL	Sulfur to P-101A/B (steam-jacketed)	DN 100 (ID 102.3)	140	10	40.9	1790	0.77	0.85	0.53	1.6e+04	14.894	0.3–1	✓
L-006	6	SL	P-101 discharge to burner (steam-jacketed)	DN 100 (ID 102.3)	140	600	40.9	1790	0.77	0.85	0.53	1.6e+04	14.894	0.3–1	✓
L-030	30	AC	P-301 discharge to T-301	DN 600 (ID 590.6)	80	488	2880.0	1800	1.62	1.62	1.62	3.1e+05	6.093	1–2	✓
L-031	31	AC	T-301 acid to IPA tank	DN 800 (ID 794.0)	124	0	2977.9	1800	0.93	0.93	0.92	2.4e+05	1.523	0.3–1	✓
L-032	32	AC	P-302 discharge to T-302	DN 450 (ID 438.2)	80	488	1800.0	1800	1.84	1.84	1.84	2.6e+05	11.039	1–2	✓
L-033	33	AC	T-302 acid to FAT tank	DN 700 (ID 692.0)	91	0	1804.1	1800	0.74	0.74	0.74	1.7e+05	1.187	0.3–1	✓
L-U01	U1	HS	Saturated steam B-101 → SH-201	DN 250 (ID 254.5)	250	4000	94.0	20	25.73	28.41	17.24	6.5e+06	36.045	20–40	✓
L-U02	U2	HS	Superheated steam to header	DN 250 (ID 254.5)	292	4000	94.0	17	29.50	32.57	19.77	6.5e+06	41.327	20–40	✓
L-U03	U3	BW	BFW to economizers	DN 150 (ID 154.1)	105	4500	150.0	955	2.34	2.34	2.34	1.3e+06	26.837	1–3	✓

8 · Rotating equipment hydraulics

K-101: model curve PI-101 = 75·ρr − 15·(Q/Qdes)² kPa (K.H0, K.aB); shaft power from the model's econ(): 4200·(Q/Qdes)^1.5 kW, the same as the Efficiency tab. Isentropic power = $P_1 V_1 \cdot k/(k-1) \cdot [(P_2/P_1)^{((k-1)/k)} - 1]$, k = 1.4, suction at the model's 90 °C and 0 kPa g.

K-101	Design	110 % rate	Turndown
Air FI-100, kNm³/h	238.5	262.2	163.1
Flow, % of design	100.0	109.9	68.4
Suction volume, m³/s	88.1	96.8	60.2
Discharge PI-101, kPa g	60.0	56.9	68.0
Shaft power (econ), kW	4200	4840	2375
Isentropic gas power, kW	4440	4661	3375
Implied isentropic efficiency, %	105.7	96.3	142.1
Isentropic discharge T, °C	141.6	139.3	147.4
Efficiency tab power, kW	4200	4840	2375

At 90 °C suction the gas needs 4440 kW, more than the 4200 kW shaft power, and the model adds no compression heat. With 45 °C suction (typical after a drying tower, assumed) the isentropic power is 3889 kW and the isentropic discharge about 90 °C: see EQ-ep-03.

Pumps

Pump	Service	Normal m³/h	Rated m³/h	ΔP kPa	Head m	Hydraulic kW	η	Shaft kW	Motor kW	Basis, NPSH
P-101A/B	Molten sulfur to F-101	22.9	25	590	33.6	3.8	0.50	7.5	9	Vertical pit pump, steam-jacketed (typical). Model supply pressure K.Psup 600 kPa across FV-103. Vapour pressure of sulfur at 140 °C is negligible; NPSHa = submergence.
P-301A/B	IPA acid circulation	1600.0	1760	488	27.7	217.1	0.78	278.3	322	Head assumed: lift 22 m + cooler 70 kPa + distributor/line 30 kPa. Vertical pump in the acid tank (typical): NPSHa = submergence; 98.5 % acid vapour pressure at 80 °C ≪ 1 kPa. FSLL-301 trip at 800 m³/h (50 %); P-301B auto-start.
P-302A/B	FAT acid circulation	1000.0	1100	488	27.7	135.7	0.78	174.0	201	Same head basis as P-301 (assumed). FSLL-302 trip at 500 m³/h (50 %); P-302B manual start only.

9 · Design margins and the operating window

Typical practice: rated flow = normal + 10 %; design pressure = max(operating + 10 %, operating + 170 kPa); design temperature = operating + 25–30 °C (here +30 °C, rounded up to 5 °C), operating = highest of the three cases. Mechanical design conditions are issued in SO3-V1-ME-EQL-001.

Item (streams)	Max op. P kPa g	Design P kPa g	Max op. T °C	Design T °C	Note
K-101 / air ducts (2, 3, 4)	68.0	238	90.0	120	blower shut-off 75 kPa g (K.H0) must be inside design P
F-101 / B-101 gas side (7–11)	49.5	220	1123.5	1155	refractory lined: shell design T set by the lining (typical 250–350 °C), not the gas
R-201 pass 1, bed 1 exit (15)	42.3	212	632.3	665	compare with TSHH-203 trip 660 °C
R-201 pass 2 (24–27)	15.2	185	442.5	475	
E-201 / E-202 / SH-201 hot side (15–19)	36.8	207	632.3	665	
T-301 / T-302 towers (21–22, 28–29)	22.7	193	190.0	225	brick-lined CS (typical)
Acid pump discharge (30, 32)	488.5	658	128.8	160	design T on the hottest acid in the circuit (31)
Sulfur pump discharge (6)	600.0	770	140.0	170	steam tracing at ~ 0.4 MPa g (typical) keeps 135–155 °C

Item (streams)	Max op. P kPa g	Design P kPa g	Max op. T °C	Design T °C	Note
B-101 steam side / HP steam	4000.0	4400	294.1	325	saturation K.Tsat 250 °C; superheated TI-207

Operating window (Operating windows tab, 12 % SO_2, design air): sulfur 20.0–44.7 t/h. Design 40.94 t/h is inside with 9.2 % margin to the top; the turndown case 28.0 t/h is inside; the 110 % case 45.0 t/h is outside (FV-101 97.1 %, PI-200 42.3 kPa g): see EQ-ep-04.

10 · Checks against the model and the other documents

32 checks: 24 ✓, 8 ✗. Each ✗ is an engineering query. The tab recomputes the same checks in the page.

#	Check	Value	Criterion	Result	Against	Query
1	HMB design case = model at snapDesign (33 streams recomputed now)	max Δṁ 0.001 %, max ΔT 0.05 K	< 0.1 % and < 0.5 K	✓	snapDesign	
2	Overall mass closure, Design	0.0022 %	|closure| < 0.1 %	✓	HMB §3	
3	Sulfur balance closure, Design	0.0000 %	|closure| < 0.1 %	✓	FIC-103, FI-309, AIT-306	
4	Overall mass closure, 110 % rate	0.0022 %	|closure| < 0.1 %	✓	HMB §3	
5	Sulfur balance closure, 110 % rate	0.0000 %	|closure| < 0.1 %	✓	FIC-103, FI-309, AIT-306	
6	Overall mass closure, Turndown	0.0022 %	|closure| < 0.1 %	✓	HMB §3	
7	Sulfur balance closure, Turndown	0.0000 %	|closure| < 0.1 %	✓	FIC-103, FI-309, AIT-306	
8	HMB stream numbers = PFD-201 numbered streams	33 rows / 33 on PFD-201	same set	✓	pf_spec.json	
9	PFD-201 flag values = HMB design values as rounded	29 of 29	all	✓	PFD tab	
10	Every BFD stream (1–21) appears in the HMB	21 of 21	21	✓	BFD tab	
11	Line velocities at design inside the typical criteria	27 of 27 lines	all	✓	§4 line list	
12	Line velocities at 110 % rate ≤ upper criterion (lines sized on the largest case)	27 of 27	all	✓	§4	
13	K-101 power = Efficiency tab, Design	4200 vs 4200 kW	±1 %	✓	EF_D.json	
14	K-101 power = Efficiency tab, 110 % rate	4840 vs 4840 kW	±1 %	✓	EF_D.json	
15	Stack SO_2 = Efficiency tab rate sweep, 110 % rate	342.8 vs 342.8 ppm	±1 %	✓	EF_D.json	
16	K-101 power = Efficiency tab, Turndown	2375 vs 2375 kW	±1 %	✓	EF_D.json	
17	Stack SO_2 = Efficiency tab rate sweep, Turndown	184.7 vs 184.7 ppm	±1 %	✓	EF_D.json	
18	Sulfur rate inside the operating window, Design	40.94 t/h in 20.0–44.7	inside	✓	Operating windows tab	
19	Sulfur rate inside the operating window, 110 % rate	45.00 t/h in 20.0–44.7	inside	✗	Operating windows tab	EQ-ep-04
20	Sulfur rate inside the operating window, Turndown	28.00 t/h in 20.0–44.7	inside	✓	Operating windows tab	
21	K-101 implied isentropic efficiency (gas power / econ power)	105.7 %	≤ 90 % (typical)	✗	econ(), PI-101	EQ-ep-03
22	K-101 discharge temperature > suction (compression heat)	stream 2 90.0 °C vs stream 1 90.0 °C	$T_2 > T_1$	✗	model	EQ-ep-03
23	B-101 gas-side duty = steam × 2361 kJ/kg (model basis)	61.64 vs 61.64 MW	±1 %	✓	BFD tab	
24	B-101 duty in the Process tab (boiling only) = HMB gas-side duty	44.8 vs 61.6 MW	±1 %	✗	Process tab	EQ-ep-01
25	Sulfur heat: model K.DHS vs Process/BFD tabs (296.8 kJ/mol)	104.6 vs 105.3 MW	±0.1 %	✗	K.DHS	EQ-ep-02
26	Converter heat release = BFD tab (34.9 MW)	34.9 MW	±0.5 %	✓	BFD tab	
27	Heat recovered (steam + superheat + BFW) = BFD/Efficiency tabs (113.4 MW)	113.4 MW	±0.5 %	✓	BFD / Efficiency tabs	
28	R-201 bed 1 exit design T ≥ TSHH-203 trip + 15 K	665 °C vs 675 °C	≥	✗	SIS I-2	EQ-ep-06
29	Air system design P ≥ K-101 shut-off pressure	238 vs 75 kPa g	≥	✓	K.H0	
30	Steam saturation K.Tsat = Tsat at 4.0 MPa g (251.8 °C, steam tables)	250.0 °C	±1 K	✗	B-101	EQ-ep-05
31	PI-200 = ΣPDI-2x4 + K.dPrest	35.0 vs 36.6 kPa	±0.5 kPa	✗	model	EQ-ep-08
32	Acid circulation (rated 110 %) above the FSLL-301/302 trip (50 %)	1760 / 1100 m³/h vs 800 / 500	>	✓	SIS I-3, I-4	

11 · Engineering queries

Id	Discipline	Subject	Finding	Proposed
EQ-ep-01	Process	B-101 duty basis	The HMB gas-side duty of B-101 is 61.6 MW (= FI-108 × 2361 kJ/kg, the model raises steam from water at 440 kJ/kg). The Process tab quotes 44.8 MW (= FI-108 × (2801 − 1087) kJ/kg, boiling	Use the gas-side duty 61.6 MW for the B-101 datasheet; state the steam-side split (sensible + boiling) on the steam-system HMB and align the Process tab

Id	Discipline	Subject	Finding	Proposed
			only). The BFW is already heated in E-203/E-204/E-205 (49.5 MW), so if the economizer water feeds the B-101 drum, the model counts the BFW preheat twice.	text.
EQ-ep-02	Process	Heat of sulfur combustion	The model furnace uses K.DHS = 295 kJ/mol (104.6 MW at design); the Process, BFD and Efficiency tabs use 296.8 kJ/mol (105.3 MW).	Adopt one value (the model's 295 kJ/mol in this HMB) and correct the other tabs, or change K.DHS and re-run.
EQ-ep-03	Process / Mechanical	K-101 power vs gas power	With the model's suction at 90.0 °C and atmospheric pressure, the isentropic gas power for 238.5 kNm³/h and 60.0 kPa is 4440 kW, above the 4200 kW shaft power in econ() (implied efficiency 106 %). The model also gives no temperature rise across K-101 (stream 2 = stream 1 = 90.0 °C); 4200 kW into the air would raise it ≈ 48.8 K.	Confirm whether 90 °C is the blower suction or discharge temperature. With 45 °C suction (typical after the drying tower, assumed) the isentropic power is 3889 kW; at 80–85 % efficiency the shaft power is ≈ 4576–4862 kW. Size the motor on that (electrical load list).
EQ-ep-04	Process	Rated capacity 110 %	The 110 % case (45 t/h) is outside the operating window 20–44.7 t/h: FV-101 97.1 % open (bound 95 %), PI-200 42.3 kPa g (alarm 42), K-101 4840 kW. The design capacity has only 9.2 % margin to the window top.	Either state the design capacity as 109 % of normal, or rate K-101 head / FV-101 Cv for 110 % flow at PI-200 ≤ 42 kPa g and confirm the converter ΔP at 110 %.
EQ-ep-05	Process / Mechanical	Steam pressure basis	K.Tsat = 250 °C is the saturation temperature at ≈ 3.98 MPa abs, while B-101 steam is specified at 4.0 MPa g (4.10 MPa abs, Tsat 251.8 °C, steam tables).	State steam pressure as gauge on all documents; use 251.8 °C (or set the drum to 3.88 MPa g) on the B-101 datasheet.
EQ-ep-06	Process / Mechanical	Bed 1 exit design temperature	The typical rule (operating + 30 °C) on the highest bed 1 exit of the three cases (632.3 °C) gives 665 °C, only 5 K above the TSHH-203 trip at 660 °C; a runaway reaches the trip before the SIS acts (5 s delay) and can overshoot.	Set the R-201 pass-1 shell / internals design temperature on the trip setting plus overshoot (e.g. ≥ 680 °C, to be confirmed by the Mechanical tab and the catalyst vendor limit).
EQ-ep-07	Process	Steam / BFW balance	BFW to the economizers FI-253 = 150 t/h but steam FI-108 = 93.99 t/h; the model treats the economizer water and the boiler as separate circuits, so the steam-drum balance (blowdown, deaerator, export) cannot be closed from the model.	Issue a utility HMB for the steam system (drum, blowdown ≈ 1–2 % typical, deaerator) with the economizer water routed to the drum.
EQ-ep-08	Process	Converter ΔP basis	PI-200 (35.0 kPa g) is computed from the cold design bed ΔP (ΣK.dPbed = 14 kPa) plus K.dPrest 21 kPa, while PDI-204…244 read the temperature-corrected ΔP (Σ = 15.6 kPa). Their sum plus K.dPrest is 36.6 kPa, not 35.0.	Use one ΔP basis for PI-200 and the PDIs; this HMB keeps the PDIs and allocates the remaining 19.4 kPa to exchangers, towers and ducts (assumed split).

12 · Assumptions

- Molten sulfur 140 °C, 1790 kg/m³, 9 mPa s; pump suction 10 kPa g (submerged pit pump); P-101 efficiency 0.5.
- Acid 98.5 % at 80 °C: 1800 kg/m³ (as the BFD tab), 5.5 mPa s, cp 1.5 kJ/kg K; SO_3 absorption heat 132 kJ/mol (Efficiency tab); FAT acid inlet 80 °C.
- Acid pump head: lift 22 m + cooler 70 kPa + distributor/line 30 kPa = 27.7 m; efficiency 0.78; motor efficiency 0.95, motor rating +10 %.
- Gas leaving T-302 80 °C (Efficiency tab). Gas viscosity of air (Sutherland), roughness 0.15 mm ducts, 0.05 mm pipes.
- Pressure profile: PI-101, PI-200, PDI-204…244 and K.Psup from the model; furnace + WHB share the model resistance K.kf (F-101 one third, assumed); PI-200 − ΣPDI (19.4 kPa) split over SH-201, E-201/E-202 (both sides), E-203, T-301, E-204, E-205, T-302 and the stack duct in fixed shares (assumed).
- Steam: 4.0 MPa g; superheated density 17.4 kg/m³, saturated 19.95 kg/m³ (steam tables, approximate); BFW 955 kg/m³ at 105 °C, 0.27 mPa s, supply 4.5 MPa g (assumed).
- Velocity criteria and design-margin rules are typical practice, not a code requirement.